

Regularized Regression as a Candidate Baseline

Scientific Question

Is a simple regularized model a reasonable baseline before a larger experiment? This note considers a synthetic, public-safe prediction example to support a narrow decision about the next comparison set.

The running example is a prediction problem with paired observations (x_i, y_i) , where $x_i \in \mathbb{R}^p$ and y_i is a continuous endpoint. The baseline estimator solves

$$\hat{\theta}_0 = \operatorname{argmin}_{\theta \in \mathbb{R}^p} \frac{1}{n} \sum_{i=1}^n (y_i - x_i^\top \theta)^2.$$

The regularized candidate uses

$$\hat{\theta}_\lambda = \operatorname{argmin}_{\theta \in \mathbb{R}^p} \left[\frac{1}{n} \sum_{i=1}^n (y_i - x_i^\top \theta)^2 + \lambda \|\theta\|_2^2 \right].$$

Evidence Design

The comparison is intentionally small. It is meant to check rendering and decision structure, not to claim a scientific discovery. We evaluate two synthetic folds, report root mean squared error, and inspect whether the regularized model is consistently lower without hiding the uncertainty.

Fold	Baseline RMSE	Regularized RMSE	Delta	Interpretation
A	1.42	1.18	-0.24	regularization helps
B	1.36	1.29	-0.07	effect is smaller
Mean	1.39	1.235	-0.155	promising but limited

The average reduction is

$$\Delta_{\text{mean}} = \frac{1}{2} \sum_{k \in \{A, B\}} (\text{RMSE}_{\lambda, k} - \text{RMSE}_{0, k}) = -0.155.$$

Interpretation

The result supports using the regularized model as a conservative baseline in a larger experiment. It does not justify a claim that the regularized model is optimal, and it does not distinguish whether the improvement comes from variance reduction, implicit feature weighting, or fold-specific noise. The advisor decision is therefore narrow: keep the regularized model in the next comparison set, but do not make it the final method.

The main caveat is that two folds are not enough to estimate variability. A larger run should include at least five folds or repeated random splits, and the report should show the distribution of fold-level differences rather than only the mean. If the next run still shows a negative mean delta and no severe outlier fold, the method is worth using as the default baseline.